

PROJECT 2

Advanced SQL Data Warehouse & Business Analytics

Central Region Sales & Profitability Analysis

An executive report on data warehouse design, KPI monitoring, and profitability drivers for the Central retail region

Business Scenario: Retail Superstore — Central Region

Scope: 2,323 order line items · Full fiscal year

Prepared: September 2026

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1. Executive Summary

The Central region generated **\$501,240 in total sales and \$39,706 in profit** over the reporting period, resulting in a **7.92% profit margin**. This indicates that strong sales volume is not translating efficiently into profitability.

The analysis identifies a clear pattern: the **Technology category generates approximately 85% of the region's total profit**, while the Furniture category posts a net loss and Office Supplies generates only a modest margin despite comparable sales volume. A closer look shows that the Furniture problem is concentrated in three specific sub-categories — **Tables, Bookcases, and Furnishings** — which collectively offset the healthy profit generated by Chairs.

This report documents the data warehouse built to support the analysis, presents the key findings with supporting figures, and closes with four targeted, data-backed recommendations to improve the region's profitability.

2. Business Scenario & Objective

A retail organization wants to transform its operational sales data into an analytical relational database for executive reporting and KPI monitoring. The Central region's raw transactional records were used as the source dataset, with the goal of answering three core executive questions:

- Which product categories and sub-categories drive profit, and which quietly erode it?
- Is overall profitability healthy, or does strong revenue mask a weak margin?
- Which customer segments and shipping methods operate most efficiently?

3. Methodology: Building the Data Warehouse

The raw operational table (central_superstore) was profiled, cleaned, and validated before being restructured into a dimensional star schema — the standard architecture for fast, reliable business intelligence reporting.

3.1 Data Quality Validation

- **Result: All columns checked for missing values**
- **Result: row_id, order_id, and customer_id checked for unresolved duplicates**
- **Result: Referential integrity confirmed between fact and dimension tables after load**

3.2 Star Schema Structure

Table	Rows	Role
fact_sales	2,323	Central fact table — one row per order line item
dim_customer	629	Customer attributes and segment
dim_product	1,310	Product category, sub-category, and name
dim_location	195	City, state, region, postal code
dim_date	1,058	Order and ship date attributes
dim_ship_mode	4	Shipping method

Table 1 — Star schema tables and row counts. Fact-table row count matches the source table exactly, confirming no data loss during load.

4. Key Performance Indicators

The following headline metrics summarize the Central region's overall commercial performance for the period.

\$501,240 Total Sales	\$39,706 Total Profit	7.92% Profit Margin	\$426.59 Avg. Order Value
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A **7.92% profit margin** indicates that revenue generation is not the region's primary constraint. The main opportunity lies in improving how sales are converted into profit through product mix, pricing, discounting, and customer-level performance. Section 5 breaks this down by category.

5. Category Performance Analysis

Sales are almost evenly split across the three product categories, but profitability diverges sharply. Technology converts revenue into profit at a much higher rate than Office Supplies, while Furniture posts a net loss on a comparable sales base.

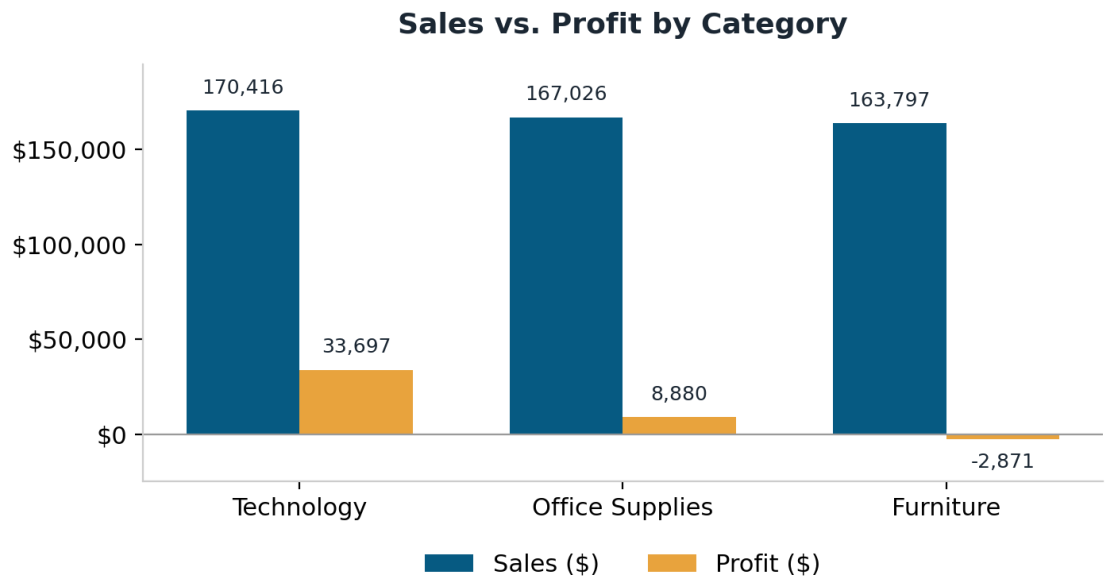


Figure 1 — Sales and profit by product category.

Category	Sales (\$)	Profit (\$)	Margin
Technology	170,416.31	33,697.43	19.77%
Office Supplies	167,026.42	8,879.98	5.32%
Furniture	163,797.16	-2,871.05	-1.75%

Table 2 — Category-level sales, profit, and margin.

- **Finding: Technology is the main profit engine of the region, contributing approximately 85% of total profit from just under a third of total sales.**
- **Finding: Furniture loses money overall despite generating sales volume nearly identical to Technology.**
- **Finding: Office Supplies generates a relatively low margin compared with Technology despite its significant contribution to total sales.**

6. Deep Dive: The Furniture Category

Furniture's category-level loss could suggest a broad structural problem, but sub-category analysis shows the issue is concentrated, not universal.

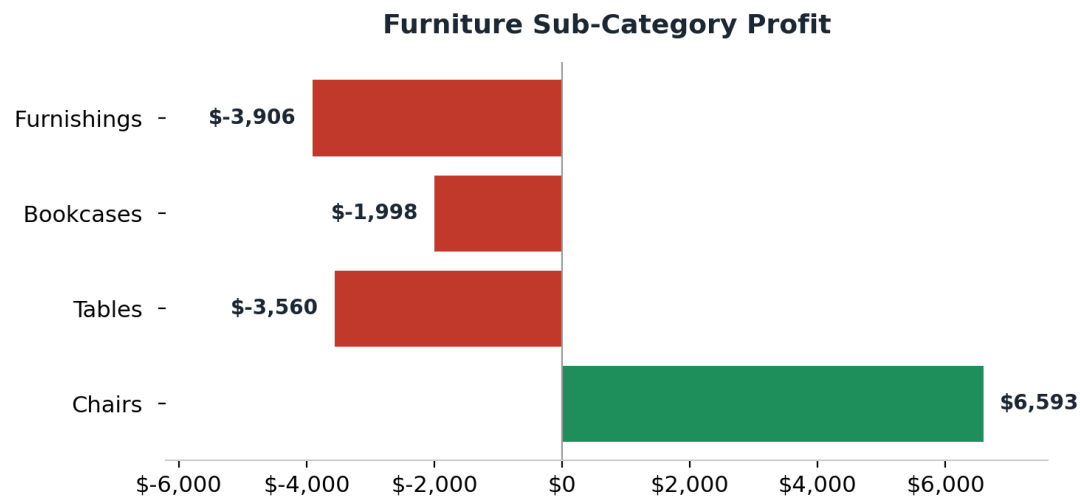


Figure 2 — Profit by Furniture sub-category.

Sub-Category	Profit (\$)
Chairs	+6,592.72
Bookcases	-1,997.90
Tables	-3,559.65
Furnishings	-3,906.22

Table 3 — Furniture sub-category profit breakdown.

- **Finding:** Chairs alone generates \$6,592.72 in profit, making it the strongest-performing Furniture sub-category in the region.
- **Finding:** Tables, Bookcases, and Furnishings together post a combined loss of \$9,463.77, more than offsetting the profit generated by Chairs and pushing the overall category into a loss.
- **Implication:** The concentration of losses in these three sub-categories suggests that pricing, discounting, or product cost should be investigated at the sub-category level rather than treating Furniture as a single underperforming category.

7. Customer Segment & Shipping Analysis

7.1 Customer Segments

Segment	Sales (\$)	Profit (\$)	Margin
Consumer	252,031.43	8,564.05	3.40%
Corporate	157,995.81	18,703.90	11.84%
Home Office	91,212.64	12,438.41	13.64%

Table 4 — Sales, profit, and margin by customer segment.

- **Finding:** Consumer is the largest segment by sales but the least efficient in terms of profit margin, converting only 3.40% of revenue into profit.
- **Finding:** Home Office is the smallest segment by revenue but has the highest profit margin, indicating stronger profitability per dollar of sales.
- **Finding:** Corporate also generates a substantially higher margin than Consumer despite having lower total sales.

7.2 Shipping Modes

Ship Mode	Sales (\$)	Profit (\$)	Avg. Shipping (days)
Standard Class	318,527.56	25,352.38	4.99
Second Class	103,550.01	9,114.84	3.33
First Class	58,746.92	3,707.27	2.32
Same Day	20,415.41	1,531.88	0.03

Table 5 — Sales, profit, and average shipping duration by shipping mode.

- **Finding:** Standard Class carries the majority of sales and profit, making it the dominant shipping mode in the region.
- **Finding:** Faster shipping modes handle smaller sales volumes and contribute less total profit, while Standard Class accounts for the largest share of commercial activity.

8. Recommendations

8.1 Prioritize the Technology category

Focus marketing and inventory availability on the highest-profit Technology sub-categories, particularly **Copiers and Phones**, to support the region's strongest profit driver.

8.2 Correct pricing on Tables, Bookcases, and Furnishings

Review discount levels, selling prices, and product costs specifically for these three sub-categories rather than the Furniture category as a whole, since Chairs already performs well and should not be affected by broad corrective actions.

8.3 Tighten discounting in Office Supplies

A **5.32% margin on \$167,026 in sales** suggests an opportunity to review discount and pricing policies within Office Supplies. A targeted review of discount thresholds can help determine whether current pricing is limiting the category's profitability.

8.4 Review Consumer segment pricing separately

As the largest segment by revenue but the lowest by margin (**3.40%**), Consumer-segment discount and pricing rules warrant a dedicated review, distinct from the more efficient Corporate and Home Office segments.

9. Conclusion

The Central region is profitable, but its profitability is concentrated: **Technology** is the primary contributor to overall profit, while **Furniture** actively reduces profitability and **Office Supplies** contributes a relatively modest margin despite substantial sales volume.

The analysis also shows that the main opportunities are concentrated in a small number of Furniture sub-categories and in the Consumer customer segment. As a result, the recommended corrective actions are targeted rather than region-wide, making them easier to implement and measure.

These actions provide practical opportunities to improve the region's profitability without requiring a broad change in its overall sales strategy.